

REFERENCES

- Holdsworth, J., Kosinski, M. and Jonker, A. 2026. *What Is a Data Warehouse?*. IBM. Available at: <https://www.ibm.com/think/topics/data-warehouse> [Accessed 10 September 2026].
- Amazon Web Services (2026) *What is a Data Lake?*, AWS. Available at: <https://aws.amazon.com/what-is/data-lake/> (Accessed: 11 September 2026).
- MongoDB (2026) *Databases vs. Data Warehouses vs. Data Lakes*, MongoDB Resources. Available at: [mongodb.com](https://www.mongodb.com/resources) (Accessed: 11 September 2026).
- David (2026) *Data Warehouse vs Data Lake vs Data Lakehouse*, Data Stack Hub Comparisons. Available at: <https://www.datastackhub.com/comparison/data-warehouse-vs-data-lake-vs-data-lakehouse/> (Accessed: 11 September 2026).
- Qlik (2026) *What is a Data Mart? Definition, Benefits, Types*, Qlik Resources. Available at: <https://www.qlik.com/us/data-warehouse/data-mart> (Accessed: 11 September 2026).
- Lu, T. (2025) *Data Mart vs Data Warehouse: a Detailed Comparison*, DataCamp Blog. Available at: datacamp.com (Accessed: 11 September 2026).
- Raveh, D., 2024. *Semi-Structured Data Explained*. CrowdStrike Cybersecurity 101. Available at: <https://www.crowdstrike.com/en-us/cybersecurity-101/cloud-security/semi-structured-data/> [Accessed 12 September 2026].

- Ademola-Shanu, W., 2024. *What is Data? Structured vs Unstructured Data Explained Simply for Beginners*. Medium. Available at: medium.com [Accessed 12 September 2026].
- Marr, B., 2021. *What's The Difference Between Structured, Semi-Structured And Unstructured Data?* Bernard Marr. Available at: <https://bernardmarr.com/whats-the-difference-between-structured-semi-structured-and-unstructured-data/> [Accessed 12 September 2026].
- Pykes, K. (2025) *Data Modeling Explained: Techniques, Examples, and Best Practices*, DataCamp. Available at: <https://www.datacamp.com/blog/data-modeling> (Accessed: 12 September 2026).
- Olin, T. (2025) *Understanding Data Modeling and types of data models*, Agile Data Engine. Available at: <https://www.agiledataengine.com/blog/data-modeling-and-data-model-types> (Accessed: 12 September 2026).
- Chia, A. (2024) *Data Mining: The Ultimate Introduction*, Splunk. Available at: https://www.splunk.com/en_us/blog/learn/data-mining.html (Accessed: 12 September 2026).
- Lee, N. (2022) *Data mining examples in the real world*, Safe Software. Available at: <https://fme.safe.com/blog/2022/09/dataminingexamples/> (Accessed: 12 September 2026).
- Knight, M., 2024. *What Is a Database Management System (DBMS)?*. Dataversity. Available at: <https://www.dataversity.net/data-concepts/what-is-a->

database-management-system-dbms/ [Accessed 12 September 2026].

- StackScale, 2023. *10 popular database management systems (DBMS)* [online]. StackScale Blog. Available from: <https://www.stackscale.com/blog/popular-database-management-systems/> [Accessed 12 September 2026].
- Wang, C. (2026). *What is ETL? A simple guide to the ETL process and how it works*. Fivetran. Available at: <https://www.fivetran.com/blog/what-is-etl> [Accessed 12 September 2026].
- PowerMetrics, 2025. *What is ETL? Extract, Transform & Load explained (with MCP context)* [online]. PowerMetrics Blog by Klipfolio. Available from: <https://www.powermetrics.app/blog/what-is-etl> [Accessed 12 September 2026].
- Amazon Web Services, 2026. *What is ETL (Extract Transform Load)?* [online]. AWS Cloud Computing Concepts Hub. Available from: <https://aws.amazon.com/what-is/etl/> [Accessed 12 September 2026].
- Chen, M. 2024. *What Is Machine Learning?* Oracle. Available at: <https://www.oracle.com/africa/artificial-intelligence/machine-learning/what-is-machine-learning/> [Accessed: 12 September 2026].
- Data Education Center, 2026. *What Are the Different Types of AI? Narrow AI, General AI, Super AI*. Support Documentation. Available at: support.site [Accessed 12 September 2026].

- Temov, J., 2026. *Big Data File Formats Explained*. Enov8 Blog. Available at: <https://www.enov8.com/blog/big-data-file-formats-untangled/> [Accessed 12 September 2026].
- Lee, F. 2026. *What is a neural network?* IBM. Available at: <https://www.ibm.com/think/topics/neural-networks> [Accessed: 12 September 2026].
- Amazon Web Services. 2026. *What's the Difference Between Machine Learning and Deep Learning?* Available at: <https://aws.amazon.com/compare/the-difference-between-machine-learning-and-deep-learning/> [Accessed: 12 September 2026]
- International Organization for Standardization (ISO). 2026. *Deep learning: The mechanics of magic*. Available at: <https://www.iso.org/artificial-intelligence/deep-learning> [Accessed: 12 September 2026].
- Syracuse University iSchool. 2025. *What Is Machine Learning? Key Concepts and Real-World Uses*. Syracuse University School of Information Studies. Available at: <https://ischool.syracuse.edu/what-is-machine-learning/> [Accessed: 12 September 2026].
- OTC Team. 2026. *Machine Learning vs Traditional Programming*. Oxford Training Centre Uk. Available at: oxfordcentre.uk [Accessed: 12 September 2026].
- Belcic, I. and Stryker, C. 2026. *What is GPT (generative pretrained transformer)?* IBM Think. Available at: [ibm.com](https://www.ibm.com/think/topics/gpt) [Accessed: 13 September 2026]

- Databricks Staff. 2026. *What are Large Language Models (LLM)?* Databricks. Available at: <https://www.databricks.com/blog/what-are-large-language-models> [Accessed: 13 September 2026].
- Toloka Team. 2026. *Difference between AI, ML, LLM, and generative AI.* Toloka AI BV. Available at: <https://toloka.ai/blog/difference-between-ai-ml-llm-and-generative-ai/> [Accessed: 13 September 2026].
- Luna, J.C. 2023. *What is an LLM? A Guide on Large Language Models and How They Work.* DataCamp. Available at: <https://www.datacamp.com/blog/what-is-an-llm-a-guide-on-large-language-models> [Accessed: 13 September 2026].
- Crabtree, M. 2024. *What is Natural Language Processing (NLP)? A Comprehensive Guide for Beginners.* DataCamp. Available at: <https://www.datacamp.com/blog/what-is-natural-language-processing> [Accessed: 13 September 2026].
- Stryker, C. and Scapicchio, M. 2026. *What is generative AI?* IBM Think. Available at: [ibm.com](https://www.ibm.com/think/topics/generative-ai) [Accessed: 13 September 2026].
- Singapore Management University (SMU) Academy. 2026. *What Is Generative AI? How It Works, Use Cases & Benefits.* SMU Academy Insights. Available at: [smu.edu.sg](https://www.smu.edu.sg) [Accessed: 13 September 2026].
- Cape Peninsula University of Technology (CPUT). 2026. *What is Generative AI?* CPUT Student AI Practice Guide. Available at: <https://www.cput.ac.za/student-ai-practice-guide/what-is-generative-ai> [Accessed: 13 September 2026].

- Snowflake. 2026. *What Is Computer Vision? Applications and Use Cases*. Snowflake Resources. Available at: snowflake.com [Accessed: 13 September 2026].
- Caballar, R.D. and Stryker, C. 2026. *What is computer vision?* IBM Think. Available at: <https://www.ibm.com/think/topics/computer-vision> [Accessed: 13 September 2026]
- Rogers, J. and Jonker, A. 2024. *What is data bias?* IBM Think. Available at: [ibm.com](https://www.ibm.com) [Accessed: 13 September 2026].
- Mercor. 2026. *Why is training data important for AI models?* Mercor Expert Resources. Available at: <https://www.mercor.com/resources/experts/training-data-importance-for-ai-models/> [Accessed: 13 September 2026].
- H2O.ai. 2026. *What Are Training Sets?* H2O.ai Wiki. Available at: <https://h2o.ai/wiki/training-sets/> [Accessed: 13 September 2026].
- Jackson, G. and Goodwin, M. 2026. *What Is an API Key?* IBM Think. Available at: <https://www.ibm.com/think/topics/api-key> [Accessed: 13 September 2026].
- Kukushkina, N. 2025. *Understanding the role of APIs in modern Software Development*. DigitalMara Blog. Available at: digitalmara.com [Accessed: 13 September 2026].
- Goodwin, M. 2026. *What Is an API (Application Programming Interface)?* IBM Think. Available at: [ibm.com](https://www.ibm.com) [Accessed: 13 September 2026].

- Amazon Web Services. 2026. *What is an API (Application Programming Interface)?* Available at: <https://aws.amazon.com/what-is/api/> [Accessed: 13 September 2026].
- IBM, 2024. *JSON transformation examples: Query*. IBM DataPower Gateway 10.6.0 Documentation. [online] Available at: [ibm.com](https://www.ibm.com) [Accessed 13 September 2026].
- ITU Online Editorial Team, 2026. *What is JSON (JavaScript Object Notation)?*. ITU Online IT Training. [online] Available at: <https://www.ituonline.com/tech-definitions/what-is-json-javascript-object-notation/> [Accessed 13 September 2026].
- Amazon Web Services (AWS), 2026. *What is JSON?*. Amazon DocumentDB. [online] Available at: <https://aws.amazon.com/documentdb/what-is-json/> [Accessed 13 September 2026].
- Amazon Web Services (AWS), 2026. *What is a RESTful API?*. [online] Available at: <https://aws.amazon.com/what-is/restful-api/> [Accessed 13 September 2026].
- Amazon Web Services. 2026. *What is a RESTful API?* Available at: <https://aws.amazon.com/what-is/restful-api/> [Accessed: 13 September 2026].
- Sv, M., 2022. *Python map() Function Explained*. Towards Data Science. Available at: <https://towardsdatascience.com/python-map-function-explained-c2bb23a7c8da/> [Accessed 13 September 2026].

- Kumar, R., 2025. *Python map() Function: A Complete Guide*. DataCamp. [online] Available at: <https://www.datacamp.com/tutorial/python-map-function> [Accessed 13 September 2026].
- Gadesha, V., 2026. *Prompt engineering techniques*. IBM Think. [online] Available at: ibm.com [Accessed 13 September 2026].
- Mitchell, T., 2025. *What is Prompt Engineering? Techniques, Examples, and Tools*. The Couchbase Blog. [online] Available at: <https://www.couchbase.com/blog/prompt-engineering/> [Accessed 13 September 2026].
- GDPR.eu, 2026. *What is GDPR, the EU's new data protection law?* [online] GDPR.eu. Available at: <https://gdpr.eu/what-is-gdpr/> [Accessed 13 September 2026].
- Alation, 2024. *Why Is Data Privacy Important? A Guide for Businesses and Individuals*. [online] Alation Blog. Available at: <https://www.alation.com/blog/why-data-privacy-is-important/> [Accessed 13 September 2026].
- Kosinski, M. and Forrest, A., 2026. *What is data privacy?*. IBM Think Topics. Available at: ibm.com [Accessed 13 September 2026].
- Ali, M., 2025. *Top 31 Python Libraries for Data Science in 2026*. DataCamp Blog. Available at: datacamp.com [Accessed 13 September 2026].
- Kleemann, G., 2023. *Why Python is Best for Data Science*. Ascendient Learning Blog. Available at: ascendientlearning.com [Accessed 13 September 2026].

- Oracle, 2026. *What is Python?*. Oracle Africa Region Developer Resources. Available at: <https://www.oracle.com/africa/developer/what-is-python-for-developers/> [Accessed 13 September 2026].
- Ashwani, K., 2024. *What is Jupyter Notebook and use cases of Jupyter Notebook?*. DevOpsSchool Blog. Available at: <https://www.devopsschool.com/blog/what-is-jupyter-notebook-and-use-cases-of-jupyter-notebook/> [Accessed 13 September 2026].
- ITU Online Editorial Team, 2026. *What is Git Branch? – ITU Online IT Training*. Available at: [ituonline.com](https://www.ituonline.com) [Accessed 13 September 2026].
- Worsley, S., 2023. *What is Git? – The Complete Guide to Git*. DataCamp Blog. Available at: datacamp.com [Accessed 13 September 2026].
- Atlassian, 2025. *What is version control / Atlassian Git Tutorial*. Atlassian Git Guides. Available at: <https://www.atlassian.com/git/tutorials/what-is-version-control> [Accessed 13 September 2026].
- Amazon Web Services, 2026. *SaaS vs PaaS vs IaaS – Types of Cloud Computing*. Available at: [amazon.com](https://aws.amazon.com) [Accessed 13 September 2026].
- Susnjara, S. and Smalley, I., 2026. *What is cloud computing?*. IBM Think. Available at: <https://www.ibm.com/think/topics/cloud-computing> [Accessed 13 September 2026].

- Edmond, S. and Crabtree, M., 2026. *Power BI vs. Tableau: Which is The Better Business Intelligence Tool in 2026?*. DataCamp Blog. Available at: [datacamp.com](https://www.datacamp.com/blog/power-bi-vs-tableau-which-one-should-you-choose) [Accessed 13 September 2026].
- Edmond, S. and Crabtree, M., 2026. *Power BI vs. Tableau: Which is The Better Business Intelligence Tool in 2026?*. DataCamp Blog. Available at: <https://www.datacamp.com/blog/power-bi-vs-tableau-which-one-should-you-choose> [Accessed 13 September 2026].
- Holdsworth, J., 2023. *What is AI bias?* [online] IBM Think Topics. Available at: <https://www.ibm.com/think/topics/ai-bias> [Accessed 13 September 2026].
- SAP, 2024. *What is AI bias? Causes, effects, and mitigation strategies.* [online] SAP Insights. Available at: <https://www.sap.com/resources/what-is-ai-bias> [Accessed 13 September 2026].
- Amazon Web Services, 2026. *Data Governance with AWS.* [online] AWS Data Lakes and Analytics. Available at: <https://aws.amazon.com/big-data/datalakes-and-analytics/data-governance/> [Accessed 13 September 2026].
- Databricks, 2026. *What Is Data Governance? A Comprehensive Guide.* [online] Databricks Discover. Available at: <https://www.databricks.com/discover/data-governance> [Accessed 13 September 2026].

- SAS Institute Inc., 2026. *AI Governance: What it is and why it matters*. [online] SAS Insights. Available at: https://www.sas.com/en_us/insights/analytics/ai-governance.html [Accessed 13 September 2026].
- Cepelak, C., 2024. *An Introduction to Data Ethics: What is the Ethical Use of Data?* [online] DataCamp Blog. Available at: <https://www.datacamp.com/blog/introduction-to-data-ethics> [Accessed 13 September 2026].